

Project Proposal: Treehouse

"From napkin sketch to production-ready web code, instantly."

1. Project Overview

Treehouse is a next-generation web creation platform that bridges the gap between raw visual ideation and functional frontend development. By merging the intuitive, drag-and-drop canvas of design tools like Canva with the conversational generative power of AI design systems (similar to Claude Design/Artifacts), Treehouse allows creators to upload rough sketches, wireframes, or reference drawings and transform them into live, interactive websites.

Furthermore, Treehouse introduces **Spatial Prompting** (circling specific regions to target precise AI edits) and a proactive **UX Copilot** that audits designs for missing product logic, ensuring users never forget critical features like authentication, error states, or responsive flows.

2. The Problem Statement

The current digital product creation workflow is fractured:

- **The Translation Gap:** Designers sketch ideas on paper, whiteboards, or tools like Figma, but translating those static visuals into functional code requires hours of tedious manual coding or vague prompts in clunky AI builders.
- **Prompt Ambiguity:** Existing AI website generators rely entirely on text prompts, making it nearly impossible to specify *where* a change should happen without writing complex, frustrating descriptions.
- **The "Happy Path" Blindspot:** Designers often focus solely on aesthetics, frequently forgetting core functional requirements (such as login states, empty dashboards, or form error handling) which causes friction later during development.

3. The Solution & Value Proposition

Treehouse solves these bottlenecks through three core pillars:

1. **Sketch-to-DOM Ingestion:** Drop any hand-drawn wireframe or UI screenshot onto an infinite canvas, and watch AI parse and convert it into a structured, live layout.
2. **Precision Spatial Prompting:** Instead of vague text commands, users draw a freehand circle or selection box around any UI element and prompt precise animations, color changes, or structural updates.
3. **The UX Copilot:** An intelligent automated auditor that analyzes the generated component tree, flags missing critical functionality (e.g., *"You designed a SaaS pricing page, but forgot a login button"*), and offers one-click resolutions.

4. Key Features & User Journey (MVP Scope)

- **Infinite Visual Workspace:** A high-performance canvas to upload, position, and scale sketches or wireframes (PNG/JPG/SVG).
- **AI Layout Generation:** Automated conversion of visual wireframes into clean HTML/Tailwind CSS components with instant live previews.
- **Spatial Lasso & Contextual Editing:** Interactive selection tools allowing users to target specific components for localized AI modifications and animation states.
- **Proactive Logic Auditing:** Real-time feedback sidebar categorizing missing product requirements, hidden states (empty/loading/error), and accessibility warnings with "Add it for me" automation.
- **Developer Export:** Clean code export (Tailwind/HTML/React snippets) and one-click deployment integrations.

5. Technology Stack Architecture

- **Frontend:** Next.js (React), TypeScript, Tailwind CSS, and an advanced canvas rendering library (such as Fabric.js or Konva.js) for spatial coordinate tracking.
- **Backend & AI Pipeline:** Python, Node.js, and multi-modal vision LLMs (e.g., Anthropic Claude / OpenAI / Gemini APIs) to map image pixels and coordinates to structured JSON component trees.
- **Preview & Sandbox:** Secure code-execution sandboxes to render live, interactive UI components dynamically.
- **Infrastructure:** PostgreSQL for user data and project states; cloud storage for user-uploaded sketches.

6. Execution Roadmap & Timeline (2-Person Team)

Estimated Time to MVP: **12 to 16 weeks (3 to 4 months)** with a lean, specialized 2-person engineering team (Frontend/Canvas Engineer + Full-Stack/AI Engineer).

- **Phase 1 (Weeks 1–4): Foundation & Canvas Core**
 - Build the workspace, infinite canvas, and spatial selection (circling/bounding box math).
- **Phase 2 (Weeks 5–8): AI Vision & Generation Engine**
 - Integrate multimodal vision models to parse sketches into layout code and render live previews.
- **Phase 3 (Weeks 9–12): The UX Copilot**
 - Develop the automated component tree analysis and build the Insight Sidebar with one-click injections.
- **Phase 4 (Weeks 13–16): Polish, Export, & Launch Prep**
 - Implement code export, optimize generation latency, perform end-to-end testing, and prepare for initial release.

7. Future Vision & Scalability (Post-MVP)

- **Real-Time Collaboration:** Multi-user multiplayer cursors (similar to Figma) allowing product teams to brainstorm and edit layouts together live.
- **Advanced Component Libraries:** Direct integration with popular design systems (Tailwind UI, Shadcn, Material-UI).
- **Full-Stack Expansion:** Expanding from frontend UI generation to automated backend schema and API scaffolding.